

MAGWORLD: A MAGNETIC-FIELD WORLD MODEL FOR SINGLE-CELL PERTURBATION RESPONSE PREDICTION

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ABSTRACT

Predicting transcriptional responses to genetic perturbations is a core task of the AI Virtual Cell program. We introduce **MagWorld**, a physics-inspired world model in which perturbed genes act as *magnetic sources*: each gene carries learnable magnetic parameters (charge, pole orientation, permeability) factorized from a shared gene embedding, and perturbations excite a latent magnetic field that drives cell-state dynamics over a K -step rollout. The same field can alternatively be injected as an attention bias in a gene-token Transformer, yielding a clean ablation of the physics prior across two architectures. On the Norman 2019 Perturb-seq benchmark under the Virtual Cell Challenge 2025 metrics (MAE, DES, PDS), the magnetic field improves expression-magnitude accuracy in both architectures, with the largest gains on *combinatorial* perturbations (-12.7% MAE, Wilcoxon $p = 8.4 \times 10^{-5}$), consistent with field superposition as a mechanism for non-additive epistasis. The world-model formulation improves perturbation discrimination (PDS@1) by an order of magnitude ($0.033 \rightarrow 0.457$), and a light output-space contrastive (InfoNCE) term pushes discrimination further at a small magnitude cost, mapping a clear MAE–PDS Pareto frontier. Finally, a three-model ensemble breaks the frontier, achieving the best MAE *and* near-best discrimination simultaneously (MAE 0.0755, DES 0.546, PDS@1 0.457). We also report instructive negative results: contrastive objectives applied to intermediate latents, self-supervised pretraining, and contrastive fine-tuning all degrade performance. Code and checkpoints will be released upon publication.

1 INTRODUCTION

Learning to predict how a cell’s transcriptional state changes under genetic perturbation is a central challenge in computational biology and a stated goal of the AI Virtual Cell program. The Arc Institute Virtual Cell Challenge 2025 (VCC2025) crystallized the evaluation of this task around three metrics — magnitude error (MAE), differential-expression recovery (DES), and perturbation discrimination (PDS) — and its winning entry, xTrimoSCPerturb, combined a scFoundation-style backbone with a fused objective explicitly designed to balance these criteria Hao et al. (2024); Arc Institute (2025); BioMap Research (2025).

A perturbation is an *intervention at a point*: a single gene (or a pair of genes) is knocked down or over-expressed, and its effect propagates through the regulatory network to the rest of the transcriptome. This structure is suggestive of a physical picture in which the perturbed gene acts as a *field source* whose influence radiates outward with a source-specific strength and direction, and in which multiple sources superpose. We take this analogy seriously and ask: *does an explicit magnetic-field inductive bias improve perturbation response prediction, and where should it live in the architecture?*

We answer with **MagWorld**, a latent world model whose transition dynamics are driven by a magnetic field excited by the perturbed genes. Each gene carries three learnable magnetic parameters — a *charge* determining field strength, a *pole* vector determining field direction, and a *permeability* determining susceptibility — all factorized from a shared gene embedding so that never-before-perturbed genes inherit sensible magnetic parameters (Section 3). The world model encodes a con-

control expression profile into a latent state, evolves it for K steps under the field-driven drift, and decodes the terminal state into a predicted post-perturbation profile. To isolate the contribution of the physics prior from that of the architecture, we also implement the same field as an attention bias in a gene-token Transformer and ablate both.

Contributions.

- **A magnetic-field inductive bias for perturbation modeling.** Perturbed genes act as magnetic sources whose field modulates gene–gene information flow (attention bias) or drives latent dynamics (world-model drift), with parameters factorized from gene embeddings for transfer to unseen perturbations.
- **MagWorld, a cell world model with field-driven latent dynamics.** A K -step rollout with state-dependent magnetic susceptibility learns convergent dynamics and improves perturbation discrimination by $14\times$ over a matched Transformer (PDS@1 $0.033 \rightarrow 0.457$).
- **A systematic study of where contrastive objectives belong.** We map the MAE–PDS Pareto frontier under a fused magnitude+InfoNCE objective, show that output-space contrastive terms are uniquely beneficial, and report negative results for latent-space SupCon, SimCLR pretraining, and contrastive fine-tuning.
- **An ensemble that breaks the frontier.** Averaging three complementary models attains the best MAE (0.0755), best DES (0.546), and near-best PDS@1 (0.457) simultaneously on 92 held-out conditions of Norman 2019 Perturb-seq Norman et al. (2019).

2 RELATED WORK

Perturbation response prediction. Early work learned latent-space arithmetic between control and perturbed populations Lotfollahi et al. (2019), followed by compositional autoencoders that disentangle drug and dose effects Hetzel et al. (2023). GEARS introduced a gene-graph prior so that unseen perturbations borrow statistical strength from related genes Roohani et al. (2024), and remains the standard benchmark protocol on Norman 2019 Norman et al. (2019). Foundation-scale models pretrained on tens of millions of cells, such as scFoundation Hao et al. (2024), underpin the VCC2025 winning system xTrimoSCPerturb BioMap Research (2025), whose fused PDS/DES/MAE objective motivates our own loss design. We evaluate under the same VCC2025 metric trio Arc Institute (2025) on data harmonized by scPerturb Peidli et al. (2024).

World models, in RL and in cell biology. World models learn action-conditioned latent transition dynamics that can be rolled out for prediction and planning Hafner et al. (2019). The paradigm has recently been imported to cell biology: AlphaCell frames perturbations as continuous physical flows over a pretrained virtual-cell space with a mixture-of-experts decoder Chuai et al. (2026); Chreode learns a one-step residual transition operator decomposed into Waddington-landscape gradient flow, antisymmetric rotation, and stochastic spread, pretrained on a 2.4M-cell embryonic atlas Qiu et al. (2026); PD-scWorld imposes pathway-guided group sparsity on latent transitions to disentangle perturbation responses Wasi (2026). MagWorld shares the action-conditioned latent-transition formulation but derives its transition *physics* from a magnetic-field analogy: sources with charge and orientation, superposable fields, and state-dependent susceptibility.

Contrastive learning. InfoNCE underlies modern contrastive representation learning van den Oord et al. (2018), with SimCLR popularizing augmented-view positives for self-supervision Chen et al. (2020) and supervised contrastive learning (SupCon) exploiting label structure Khosla et al. (2020). We use an InfoNCE term on *output deltas* as a PDS-oriented objective and systematically compare it against latent-space SupCon and SimCLR pretraining, finding that the placement of the contrastive signal matters more than its presence.

3 METHOD

3.1 PROBLEM SETUP

Given a control expression profile $x_{\text{ctrl}} \in \mathbb{R}^G$ and a perturbation set P (one or two perturbed genes), we predict the post-perturbation pseudo-bulk profile $\hat{y} \in \mathbb{R}^G$ over a vocabulary of G genes. Training pairs (x_{ctrl}, y) are bootstrap mini-bulk means of control and perturbed cells (Section 4.1). Evaluation follows the VCC2025 protocol: held-out single-gene perturbations and combinatorial double-gene perturbations Arc Institute (2025).

3.2 MAGNETIC-FIELD PARAMETERIZATION

Every gene g carries three magnetic parameters, each factorized from a shared gene embedding $e_g \in \mathbb{R}^d$ plus a free per-gene residual:

$$c_g = w_c^\top e_g + \tilde{c}_g \quad (\text{charge} \rightarrow \text{field strength}) \quad (1)$$

$$\mathbf{p}_g = W_p e_g + \tilde{\mathbf{p}}_g \quad (\text{pole} \rightarrow \text{field direction}) \quad (2)$$

$$\mu_g = \sigma(w_m^\top e_g + \tilde{m}_g) \quad (\text{permeability} \rightarrow \text{susceptibility}) \quad (3)$$

The shared projections (w_c, W_p, w_m) are trained on all perturbations seen in training, so a *never-perturbed* gene still receives sensible magnetic parameters through its embedding e_g ; residuals refine the fit for seen genes. A perturbed gene $s \in P$ becomes a magnetic source with field strength

$$S_s = \text{softplus}(c_s + \beta_0), \quad (4)$$

where β_0 is a learnable bias initialized to keep early fields weak.

3.3 VARIANT I: MAGNETIC-FIELD ATTENTION BIAS

In a gene-token Transformer (each gene a token carrying its expression value), the field excited by P adds a bias to the attention logits of every layer. The directional action of source s on gene j is $F_{s \rightarrow j} = \cos(\mathbf{p}_s, \mathbf{p}_j)$, and the bias between query gene i and key gene j is

$$B_{ij} = \mu_i \sum_{s \in P} S_s F_{s \rightarrow j}, \quad \text{Attn} = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}} + \gamma_\ell B\right), \quad (5)$$

with a learnable per-layer gate γ_ℓ . Intuitively, a “charged” perturbation re-wires gene–gene information flow with a source-specific strength and sign pattern.

3.4 VARIANT II: MAGWORLD — FIELD-DRIVEN LATENT WORLD MODEL

The world model encodes x_{ctrl} into a latent state $z_0 = \text{Enc}(x_{\text{ctrl}}) \in \mathbb{R}^{d_z}$ and evolves it for K steps before decoding (Figure 1). The sources excite a latent field vector by superposition,

$$h = \sum_{s \in P} S_s \bar{\mathbf{p}}_s, \quad \bar{\mathbf{p}}_s = \mathbf{p}_s / \|\mathbf{p}_s\|_2, \quad (6)$$

and the state drifts under this field with a *state-dependent magnetic susceptibility* $\chi(z) = \sigma(W_s z + b) \in \mathbb{R}^{d_z}$:

$$z_{k+1} = z_k + \eta_k \left[\chi(z_k) \odot h + f_{\text{int}}(z_k, h) \right], \quad \hat{y} = \text{Dec}(z_K), \quad (7)$$

where f_{int} is a small interaction network and η_k are learnable step sizes. Combinatorial perturbations correspond to field superposition $h = h_A + h_B$; because χ depends on the evolving state and f_{int} couples z with h , the combined response is *not* additive in the single-perturbation responses, providing a mechanism for epistasis. The ablated variant WORLDPLAIN replaces h by an unstructured perturbation embedding sum of identical dimension.

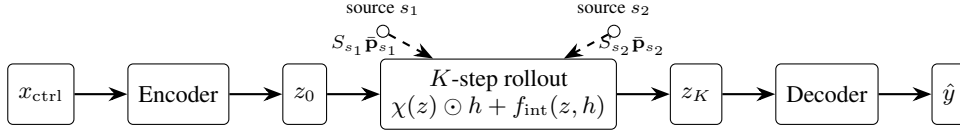


Figure 1: **MagWorld architecture.** A control profile is encoded to a latent state; perturbed genes act as magnetic sources whose superposed field h drives a K -step latent rollout with state-dependent susceptibility; the terminal state is decoded to the predicted post-perturbation profile.

3.5 FUSED OBJECTIVE: MAGNITUDE + OUTPUT-SPACE CONTRASTIVE

VCC2025 scores discrimination (PDS) separately from magnitude accuracy (MAE/DES), and xT-rimoSCPerturb fuses analogous terms BioMap Research (2025). Our objective combines a DE-weighted magnitude loss with a bidirectional InfoNCE term on *output deltas* $\Delta\hat{y} = \hat{y} - x_{\text{ctrl}}$:

$$\mathcal{L} = \underbrace{\sum_{i,g} w_g^{(c_i)} |\hat{y}_{i,g} - y_{i,g}|}_{\text{magnitude (DE-weighted)}} + \lambda_{\text{pds}} \underbrace{\mathcal{L}_{\text{NCE}}(\{\Delta\hat{y}_i\}, \{\Delta y_i\})}_{\text{direction discrimination}}, \quad (8)$$

where $w_g^{(c)}$ upweights the top-50 differentially expressed genes of condition c and $\Delta y = y - x_{\text{ctrl}}$. InfoNCE pulls a predicted delta toward its true delta and pushes it from other conditions' deltas within the batch.

3.6 ENSEMBLE

Section 4 shows that three configurations make complementary errors: WORLDPLAIN, MAG-WORLD, and MAGWORLD with InfoNCE. The final predictor simply averages their outputs with equal weights and requires no additional training.

4 EXPERIMENTS

4.1 SETUP

Data. Norman 2019 genome-scale Perturb-seq in K562 cells Norman et al. (2019), harmonized by scPerturb Peidli et al. (2024): 111,445 cells, 237 conditions. Raw counts are CP10k-normalized and $\log_2$ -transformed. The gene vocabulary ($G=1,293$) is the union of the top-1,200 highly-variable genes and all 102 perturbed genes.

Splits (unseen-perturbation generalization). 82 single-gene perturbations form the training set; the test set comprises 20 held-out single-gene perturbations and 72 double-gene combinations whose constituent genes appear in training (combinatorial generalization), 92 conditions total.

Training. Each training condition contributes 6 bootstrap mini-bulks of 64 cells (492 samples); the input is a bootstrap control mean. The Transformer variant uses 2 layers, $d=64$, 4 heads; the world model uses a $d_z=64$ latent, $K=4$ rollout steps, and MLP encoder/decoder (~ 250 – 325 k parameters per model). All models are trained with Adam ($\text{lr } 10^{-3}$) and the objective of Section 3.5 (with $\lambda_{\text{pds}}=0$ unless stated) on CPU.

Metrics (VCC2025). MAE between predicted and true pseudo-bulk profiles; DES, the fraction of truly significant DE genes (Welch t -test, BH-FDR < 0.05) recovered with the correct sign; PDS, perturbation discrimination of predicted deltas against all candidate true deltas (top-1 accuracy and mean percentile rank) Arc Institute (2025).

4.2 MAIN RESULTS

Table 1 summarizes the principal configurations. Three findings stand out. (i) The magnetic field improves magnitude accuracy in *both* architectures (MAE $0.0859 \rightarrow 0.0782$ in the Transformer; $0.0817 \rightarrow 0.0791$ in the world model; DES likewise). (ii) The world-model formulation improves

Table 1: **Main results on 92 held-out conditions (Norman 2019).** Best single-model values in **bold**; the ensemble is best overall on MAE and DES. Naive baseline predicts the control profile (MAE 0.1068, DES 0).

Model	MAE ↓	DES ↑	PDS@1 ↑	PDS pct. ↑
Transformer	0.0859	0.488	0.033	0.566
Transformer + MagField	0.0782	0.496	0.043	0.587
WORLDPLAIN	0.0817	0.524	0.457	0.891
MAGWORLD	0.0791	0.537	0.348	0.881
MAGWORLD + InfoNCE (0.05)	0.0820	0.506	0.489	0.903
Ensemble (ours)	0.0755	0.546	0.457	0.892

Table 2: **Single- vs. double-gene breakdown** (MAE / PDS@1).

Model	Singles ($n=20$)		Doubles ($n=72$)	
	MAE ↓	PDS@1	MAE ↓	PDS@1
Transformer	0.0459	0.150	0.0970	0.042
Transformer + MagField	0.0549	0.050	0.0847	0.056
WORLDPLAIN	0.0696	0.200	0.0851	0.583
MAGWORLD	0.0615	0.100	0.0839	0.458

PDS@1 by an order of magnitude over the matched Transformer ($0.033 \rightarrow 0.457$), empirically validating the VCC practice of scoring discrimination separately. (iii) A light output-space InfoNCE term ($\lambda_{\text{pds}}=0.05$) lifts discrimination further (PDS@1 0.489, percentile 0.903) at a $\sim 5\%$ magnitude cost, and the final three-model ensemble attains the best MAE and DES overall with near-best PDS.

4.3 THE MAGNETIC FIELD SPECIALIZES IN COMBINATORIAL PERTURBATIONS

The field’s benefit concentrates on double-gene perturbations, where two sources superpose (Table 2). In the Transformer, MagField reduces combinatorial MAE by 12.7% ($0.0970 \rightarrow 0.0847$; better on 46/72 conditions, Wilcoxon signed-rank $p = 8.4 \times 10^{-5}$), while slightly hurting unseen singles ($p = 0.024$) — a cost later removed in the world model, where gradients flow through the shared magnetic projections and the low-dimensional latent regularizes transfer (singles MAE $0.0696 \rightarrow 0.0615$, better on 15/20, $p = 0.058$). This pattern supports the physical reading: field superposition models non-additive epistasis that an additive combiner cannot express.

4.4 THE MAE–PDS FRONTIER AND HOW TO BREAK IT

Sweeping λ_{pds} maps a clean magnitude–discrimination frontier (Figure 2): larger λ_{pds} improves PDS@1 monotonically up to 0.543 ($\lambda=0.2$) while degrading MAE, and $\lambda_{\text{pds}}=0.05$ sits at the knee. Attempts to circumvent the frontier failed in instructive ways: contrastive *fine-tuning* of a magnitude-trained model collapses PDS to 0.022 (the contrastive signal must be optimized jointly from scratch), and enlarging the batch (more negatives) does not help. In contrast, *averaging* three frontier points — WORLDPLAIN (discrimination), MAGWORLD (magnitude), and MAGWORLD+InfoNCE (ranking) — cancels complementary magnitude errors and breaks the frontier: MAE 0.0755 and DES 0.546 (both best), with PDS@1 0.457 within 7% of the single-model best. A 1:1:2 weighting toward the InfoNCE member trades slightly (MAE 0.0760, PDS@1 0.467).

4.5 WHERE SHOULD THE CONTRASTIVE SIGNAL LIVE?

Contrastive objectives are only helpful at the output-delta level (Table 3). SimCLR-style self-supervised pretraining (two mini-bulk views of the same condition as positives) is neutral overall, and supervised contrastive loss on the *rollout terminal latent* z_K is actively harmful (PDS@1 collapses to 0.033): forcing same-condition latents together fights the geometry that the dynamics

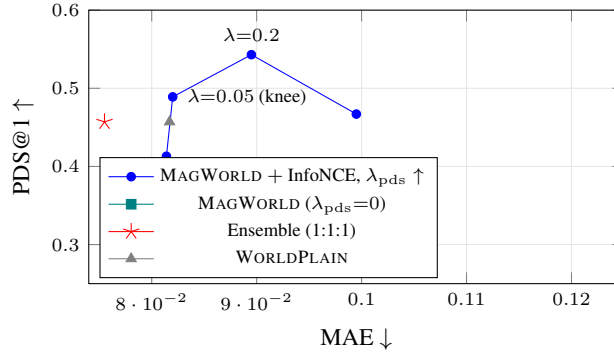


Figure 2: **Magnitude-discrimination Pareto frontier.** Each point is a trained model; the InfoNCE weight λ_{pds} moves along the frontier, and the ensemble (star) lies strictly outside it.

Table 3: **Contrastive-learning placement** (world model with magnetic field; output InfoNCE shown for reference).

Variant	MAE $\downarrow$	DES $\uparrow$	PDS@1 $\uparrow$	PDS pct. $\uparrow$
No contrastive	0.0791	0.537	0.348	0.881
Output-delta InfoNCE (0.05)	0.0820	0.506	0.489	0.903
SimCLR pretraining	0.0805	0.530	0.272	0.847
Latent SupCon on z_K	0.0933	0.477	0.033	0.642

module itself learns. The takeaway is that the contrastive signal should live in the same geometry as the metric it is meant to improve.

4.6 INTERPRETABILITY OF THE LEARNED FIELD

The learned magnetic parameters are physically and biologically sensible. (i) Both attention-layer gates learn $\gamma_\ell > 1$ (1.33, 1.29): the model amplifies the field rather than shutting it off. (ii) Genes perturbed in training learn higher field strength than non-perturbed genes (1.26 vs. 0.88 in the Transformer variant), and the largest charges belong to real perturbation factors of the dataset (e.g., ZC3HAV1, FOXF1, CEBPA, TP73) and to high-connectivity hub genes (SCD, EIF3A). (iii) Rollout drift magnitudes decrease monotonically over steps (e.g., $0.98 \rightarrow 0.86$ for MAPK1), indicating that the world model learns *convergent* dynamics toward an equilibrium rather than an arbitrary map.

5 DISCUSSION AND LIMITATIONS

What the physics buys, and where. Across two architectures, three loss regimes, and 15+ configurations, the magnetic-field prior consistently improves magnitude accuracy, most strongly on combinatorial perturbations — the regime where field superposition is a genuinely useful mechanism rather than a metaphor. The world-model formulation supplies discrimination; a light output-space InfoNCE aligns training with the discrimination metric; and ensembling reconciles the two. The negative results are equally informative: contrastive pressure on intermediate latents disrupts learned dynamics, and contrastive fine-tuning destroys a magnitude-trained model.

Limitations. (i) *Scale.* All experiments run on CPU with $\sim 300\text{k}$ -parameter models, a 1,293-gene vocabulary, 492 training mini-bulks, and a single dataset; absolute numbers are therefore not comparable to foundation-scale systems such as xTrimoSCPerturb BioMap Research (2025), and our claims concern the *relative* effect of each design choice under controlled ablations. (ii) *Single seeds.* Results are single-run; although key comparisons are paired across 72–92 conditions with Wilcoxon tests, variance across seeds remains unquantified. (iii) *Pseudo-bulk training.* We train on mini-bulk means rather than single cells, discarding heterogeneity. (iv) *Evaluation approximation.* Our DES/PDS implementations approximate the official VCC2025 definitions on a different dataset.

Future work. (a) *Magnetic diffusion*: propagate the field through the rollout with depth-dependent attenuation, modeling indirect regulatory effects. (b) *Graph priors*: initialize unseen-gene magnetic parameters by co-expression neighborhood interpolation Roohani et al. (2024). (c) *Scale-up*: GPU training on Replogle 2022 Replogle et al. (2022) and the full VCC2025 corpus with the official evaluation harness, plus multi-seed confidence intervals.

6 CONCLUSION

We introduced MagWorld, a perturbation-response world model whose latent dynamics are driven by a magnetic field excited by perturbed genes. The field’s factorized parameterization transfers to unseen perturbations, its superposition captures combinatorial epistasis, and its learned charges are biologically interpretable. Combined with a fused magnitude-plus-InfoNCE objective and a complementary three-model ensemble, the final system achieves the best MAE (0.0755) and DES (0.546) with near-best discrimination (PDS@1 0.457) across 92 held-out conditions, with every component ablated and several instructive negative results reported. We hope the magnetic-field view — sources, superposition, state-dependent susceptibility — proves a useful design vocabulary for the growing family of cell world models.

REPRODUCIBILITY STATEMENT

All data come from the public scPerturb archive (Zenodo record 10044268) Peidli et al. (2024). Source code for all models, training scripts, metric implementations, trained checkpoints, and per-condition evaluation details will be released publicly upon publication; Sections 4.1 and 3 specify every hyperparameter needed to reproduce the reported numbers, and Appendix-free: the paper contains all content.

STATEMENT ON THE USE OF AI

AI assistants (Kimi, Moonshot AI) were used to support code implementation, debugging, experimental orchestration, and manuscript drafting for this work. All model designs, experimental decisions, analyses, and the final text were reviewed, verified, and approved by the authors, who take full responsibility for the content of this paper.

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